

Personal Information

🏠 Oxford, United Kingdom

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Education

2025– **PhD in Computer Science**, *University of Oxford*, UK

Reinforcement & multi-agent learning theory. AI safety & alignment. Supervised by Jakob Foerster.

2018–2025 **Independent Researcher & Contemporary Dancer**, France, Spain, UK

Training and performing as a professional contemporary dancer. Research in spare time leading to 7 publications at ICML, NIPS, ICLR, JMLR.

2017–2018 **MSc in Mathematics and Foundations of Computer Science**, *University of Oxford*, UK

Ranked 1st, average of 94.4%. Thesis: *Stability and Exploitation in Differentiable Games*.

2013–2017 **Master of Mathematics**, *Durham University*, UK

Ranked 1st, average of 93%. Thesis: *Algebraic Features of Multiple Zeta Values*.

Publications

2026	SafeSwarm: A Multi-Agent Misalignment Benchmark	[preprint]
2026	Inverting the Bellman Equation: From Q-Values to World Models	[preprint]
2026	Procedural Generation of Algorithm Discovery Tasks in Machine Learning	[ICML]
2026	Evolution Strategies at the Hyperscale	[ICML]
2026	Goal-Conditioned Agents that Learn Everything All at Once	[ICML]
2025	An Optimisation Framework for Unsupervised Environment Design	[RLC]
2024	Tight Gradient Bounds for Parameterized Quantum Circuits	[QJ]
2023	Adversarial Cheap Talk	[ICML]
2022	Discovered Policy Optimisation	[NIPS]
2022	COLA: Consistent Learning with Opponent-Learning Awareness	[ICML]
2021	On the Impossibility of Global Convergence in Multi-Loss Optimization	[ICLR]
2020	Ridge Rider: Diverse Solutions by Following Eigenvectors of the Hessian	[NIPS]
2019	Differentiable Game Mechanics	[JMLR]
2019	Stable Opponent Shaping in Differentiable Games	[ICLR]

Awards

2026 **Cooperative AI PhD Fellowship** awarded for 3 years.

2018 **Best Thesis** in Oxford Computer Science (98%, Richard Bird Prize).

2017 **Best Thesis** in Durham Mathematics (92%, Iain MacPhee Memorial Prize).

2017 **Best Student** in Pure Mathematics (93%, Tony Corner Prize).



Experience

- 2026 **MATS (ML Alignment & Theory Scholarship)**, *Berkeley*, United States
Research scholarship to work on extracting world models from model-free agents.
- 2022 **Quantum Computing Research**, *IBM Research*, Zürich, Switzerland
Research internship in gradient bounds for parameterized quantum circuits.
- 2019 **Combinatorial ML Research**, *Giotto AI*, Lausanne, Switzerland
Research scientist developing predictive, industry-tailored models using combinatorial ML.
- 2017 **Research Internship**, *Institute of Pure and Advanced Mathematics*, UCLA, United States
3-month research project in acoustics, speech and signal processing.
- 2016 **Summer School**, *London Mathematical Society*, UK
Advanced courses and seminars in pure mathematics. Nominated and funded by the LMS.
- 2015 **Research Internship**, *Department of Mathematics*, Durham University, UK
3-month research project on the stochastic dynamics of microtubule growth.

Research Code

- 2026 **Inverting the Bellman Equation: From Q-Values to World Models**, JAX.
- 2021 **Global Convergence in Differentiable Games**, PyTorch.
- 2020 **Stable Opponent Shaping**, PyTorch.

